WWW.IJRPR.COM

International Journal of Research Publication and Reviews

Journal Homepage: www.ijrpr.com ISSN 2582-7421

Impact of Economic Policy Uncertainty and Market Volatility on Equity, Exchange Rate and Commodity Markets: Evidence from India Using Time Series Models

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ABSTRACT

This paper empirically tests how Economic Policy Uncertainty (EPU) and market volatility (India VIX) affect three systemically important Indian financial markets the equity market (Nifty 50), the foreign exchange market (USD/INR), and the commodity market (Brent crude oil) using 121 monthly observations between March 2016 and March 2026. The study uses an eight-step econometric model with ARDL bounds testing, GARCH(1,1), EGARCH(1,1), and Granger causality analysis to conclude that all the variables are I(0), there is no long-run cointegration between any two markets and that the impacts of uncertainty are short-term in nature. India VIX Granger-causes Nifty 50 returns (F=6.79, p=0.010) and EPU Granger-causes crude oil returns (F=4.22, p=0.042). The volatility persistence is also found to be much higher across the different asset classes (USD/INR: 0.9932; crude oil: 0.6403; Nifty 50: 0.4644). The results are a challenge to the assumption of homogeneous transmission of uncertainty and provide implications that can be taken to practice in portfolio management, FX hedging, energy planning, and regulatory stress-testing.

Keywords: Economic Policy Uncertainty, India VIX, Nifty 50, USD/INR, Crude Oil, GARCH, ARDL, Granger Causality, Financial Market Volatility, Emerging Markets, India.

1. INTRODUCTION

Financial markets are perpetually shaped by uncertainty, whether it is policy-driven, geopolitical or shocking to the macro economy. In an emerging economy such as India, Economic Policy Uncertainty (EPU) has a greater weight over its importance due to the reliance of the country on international capital flows, imported energy and the developmental policy interventions. With uncertainty obscuring the prospects of investors and companies, it has a direct effect on risk perception, capital allocation and eventually the valuation of financial assets in the equity, foreign exchange and commodity markets. The last ten years have been marked by a sequence of disruptive and transformative policy events in India, which have been spreading through the Indian financial ecosystem. The November 2016 demonetisation caused a sudden withdrawal of about 86% of the money in circulation, causing a severe liquidity shock that shook the value of equities and caused currency volatility. The 2020 COVID-19 pandemic was a macro-financial shock of an unprecedented scale, with nation-wide lockdowns breaking supply chains, pushing fiscal policy, and pushing investor sentiment towards extreme pessimism - Nifty 50 has fallen over 38% in a matter of weeks, India VIX has reached highs of 80, and crude oil prices went negative in global markets. More recently, the 2022 Russia-Ukraine geopolitical dispute has caused disruption to global energy and food supply chains, putting inflationary pressure on the import-dependent Indian economy, and leading to the new wave of currency and commodity market volatility.

Market volatility, particularly as captured by the India VIX - a forward-looking implied volatility measure derived from Nifty 50 options prices, is a real-time gauge of how scared and risk-averse investors are. In contrast to EPU, which indicates the policy-oriented aspect of uncertainty through newspaper coverage, VIX sums up the aggregate on the expectation of the participants in the options market and adapts immediately to new information. These two constructs EPU, a policy-oriented, backward-looking indicator and VIX, a market-implied, forward-looking indicator, together give a more holistic view of the way uncertainty spreads through financial systems.

The Nifty 50 index, representing the Indian equity market, happens to be quite vulnerable to both local and international uncertainty. Corporate earnings expectations are weakened by policy uncertainty and increase the required returns, reducing equity values. The USD/INR foreign exchange rate is reactive to the dynamics of capital flows, interest rate differentials, and demand of safe-haven currency; all of which is heightened in times of uncertainty. The commodity market, namely Brent crude oil, is an indicator of the global supply-demand factors and geopolitical risk, and India is exceptionally vulnerable due to being the third-largest crude oil importer in the world.

Although the academic and practical importance of the interaction between EPU and VIX with these markets is great, the current literature is disjointed. The vast majority of these markets are analysed in isolation, global uncertainty proxies are used instead of India-specific EPU data, and the non-normal, fat-tailed and heteroskedastic properties of the financial returns series are modelled using linear OLS frameworks, which are methodologically weak in capturing these properties. This research paper fills these gaps by using a rigorous multi-step econometric model that includes GARCH-family volatility models, ARDL cointegration test, and a VAR-based Granger causality analysis to test these variables to equity, foreign exchange, and commodity markets using India specific data between March 2016 and March 2026.

The evidence of the rest of this paper is organized in the following way: Section 2 is a literature review; Section 3 is an outline of scope and objectives of the research; Section 4 is a description of data and econometric methodology; Section 5 is a presentation of the empirical results and its interpretation; Section 6 is a discussion of the results in the theoretical and practical context; and finally, Section 7 contains policy implications, limitations, and future

2. LITERATURE REVIEW

The overlap of economic policy uncertainty and financial market behaviour has received significant academic interest, especially after the groundbreaking contribution of Baker, Bloom, and Davis (2016) who proposed text-mining-derived EPU indices as quantitative measures of policy-related uncertainty. The past ten years have seen a surge in academic research that empirically tests the EPU-market nexus in an ever more sophisticated set of econometric instruments and in a wide range of different economies and asset types.

2.1 EPU and Stock Market Performance.

Kaveh-Yazdy and Zarifzadeh (2023) presented a thorough review of the existing EPU measurement methods and identified three types and showed that text-mining methods, i.e., the tracking of policy-related words in news and financial information, are the most popular and proven ones. Their exegesis highlights the significance of proper measurement of EPU to sound economic forecasting since improperly calibrated uncertainty indices may give erroneous inferences regarding investment, unemployment, and recessionary pressures.

Kalra and Gupta (2023) directly present India-specific evidence of the equity-EPU nexus based on both linear and logarithmic regression to determine that EPU has a statistically significant negative correlation with Nifty returns and a positive correlation with stock market volatility. Importantly, their sectoral analysis shows the heterogeneity of sensitivity: the cyclical industries like automobiles are more negatively affected by the EPU impact than defensive industries like FMCG and pharmaceuticals. These results confirm EPU is a strong predictor in equity market models, and indicate that sector-specific risk management approaches may be justified at times of high policy uncertainty.

Younis et al. (2024) generalize this analysis to Chinese equity markets with wavelet-based quantile-on-quantile tools and find that the impacts of EPU and Trade Policy Uncertainty (TPU) are sector-specific and time-scale varying - most sectors are suppressed by short-run EPU shocks, whereas the banking, healthcare, and pharmaceutical sectors can be catalyzed by government stimulus Ghani and Ghani (2023) examine the predictive ability of multiple EPU indices (domestic, US, China, UK) on Pakistani stock market volatility with GARCH-MIDAS, and find the US EPU index to be the most powerful cross-border channel of uncertainty transmission, whereas the domestic EPU is of weaker predictive value - a result that may be indicative of the pre-eminent role

2.2 EPU, VIX and Foreign Exchange Markets.

Rani et al. (2025) consider the aggregate effect of both domestic and international EPU on the Indian stock market and exchange rate volatility throughout 20002021 and use the US EPU as a proxy of uncertainty spillovers across the globe. They find evidence that domestic policy changes and international uncertainty transmissions both affect Indian equity and FX volatility, and that the magnified impact of both is observed in the Global Financial Crisis (20082009) and in the COVID-19 pandemic through the use of the dummy variables. Sharma (2019) examines the effects of the exchange rate volatility on Indian export performance based on the commodity-based panel data, and finds that exchange rate uncertainty in the long-term is significantly detrimental, particularly the long-term exchange rate uncertainty, which has direct implications on the hedging strategies of Indian exporters.

The Russia-Ukraine conflict is used by Hossain, Masum, and Saadi (2024) as a natural experiment to avoid the scope of the study on the impact of geopolitical risks on the foreign exchange markets of different countries around the world. Their results indicate abrupt exchange rate depreciations - particularly in energy-intensive countries - due to increased EPU, risk aversion on the part of investors and commodity markets disturbances. Oil and gas price rises, coupled with dislocation in agricultural markets, produced compounding inflationary pressures and capital flight in the importing economies. Their work brings to light the fact that geopolitical shocks may act as concurrent EPU and VIX shocks that spread throughout FX, equity and commodity markets in a correlated and reinforcing fashion.

2.3 Uncertainty Transmission and Commodity Markets.

In a study by Sreenu (2022), the authors explore how Indian aggregate and sectoral stock returns are affected by crude oil price uncertainty, based on the Oil Price Volatility Index (OPVI). The authors use a Quantile Regression Model (QRM) to investigate the relationship between these two variables. His results reveal that the effects of OPVI changes have a materially negative effect on the returns on Indian equities, where down price shocks have larger

effects than up movements. These effects were moderated in part by the 2012 oil price deregulation reform, and it was proved that the architecture of domestic policy may mediate the international commodity price pass-through.

Dehghanzadeh Shahabad and Balcilar (2022) discuss the dynamic ARDL model relationships between oil, natural gas, and gold prices and India EPU 2003-2020. Their findings are that gold prices increase EPU (positive relationship) but industrial production decreases it (negative relationship). US/EU EPU have a positive transmission to Indian EPU, and Chinese EPU does not - highlighting India's relative integration with Western as opposed to Asian policy space. Their discovery that gold is a safe-haven in times of EPU has a direct application to Indian portfolio management in uncertain conditions.

The article by Soni, Nandan, and Chatnani (2023) explores the co-relation between EPU and Indian oil, stock, and gold markets using wavelet based coherence and quantile-on-quantile regression between November 2005 and March 2022. Their discussion shows that Indian crude oil buyers are highly hedged against the domestic effects of EPU in the short-run and also in the medium-term, but have serious difficulty in long run pricing. EPU also lowers stock market returns, whereas gold is a useful proxy of economic distortions and inflation - once again confirming its safe-haven status.

2.4 Cross-Market Spillovers and Volatility Dynamics.

Lakshmanasamy (2022) uses the GARCH estimation model to analyze the volatility transmission between the crude oil prices, exchange rates, and the BSE Sensex in the period between 2006 and 2019. His results affirm the existence of two-way volatility transmission mechanisms: the volatility of the BSE Sensex is caused by changes in the price of crude oil, and the volatility of the equity market is aggravated by fluctuations in the exchange rate. Market turbulence is further exacerbated by internal stock market shock due to endogenous feedback loops.

Lu et al. (2023) examine the effect of the COVID-19 pandemic on volatility connectedness between the Indian equity market and six major commodities markets with the help of a TVP-VAR model and wavelet coherence. They find that cross-market volatility spillovers have dramatically increased after COVID, and that cross-market hedging (using copper and crude oil) is a useful tool in risk-adjusted portfolio optimization by Indian equity investors who are uncertain about the future.

Kumar Mahalik, Acharya, and Suresh Babu (2014) analyze the price discovery and the VECM volatility spillover effect between the Indian spot and futures commodity markets by applying the VECM and the bivariate EGARCH models. Their findings indicate that agriculture, energy and aggregate commodity future markets are the most important in the transmission of spot prices, and energy futures are the main source of volatility to commodity spot markets. This publication offers the background information about the role of the futures market in the formation of the price expectations in commodities in India.

Siddiqui and Roy (2019) analyze return and volatility spillovers between Nifty 50, WTI crude oil, gold, and exchange rate using VARMA-BEKK-GARCH and DCC-GARCH models between April 2014 and March 2018. They detect both Nifty-WTI crude two-way spillovers and WTI-gold two-way spillovers and DCC-GARCH analysis proves that gold hedges Indian equity investors better, especially at times when the equity market is under stress.

2.5 Research Gaps Identified

Although the above literature is very insightful, there are a number of important gaps. To begin with, no study has examined the EPU-VIX nexus concurrently in the equity, foreign exchange, and commodity markets in a single Indian empirical study, thus being unable to compare the levels of transmission between asset classes. Second, much of the research on India is based on US or worldwide indices of EPU or works with pre-COVID-19 and Russian-Ukraine crises data, unexploring vital recent dynamics. Third, there is a dearth of literature on the use of GARCH-family models using India-specific EPU data, with much of the existing literature based on linear OLS models that are obviously broken by non-normality in Indian financial returns data. Fourth, the comparative cross-market volatility persistence analysis, i.e., the period and strength of volatility shocks in equities, FX, and commodities is a major gap in the Indian EPU literature. All four gaps are directly covered in this study.

3. SCOPE OF THE STUDY

The current research is limited to examining the concomitant effect of Economic Policy Uncertainty (EPU) and Market Volatility (India VIX) on three of the systemically relevant Indian financial markets, the equity market, the foreign exchange market, and the commodity market. Its geographic focus is national, with India being the sole unit of analysis- a decision driven by the fact that India is one of the fastest-growing emerging market economies in the world and is becoming more and more integrated into the world financial systems and, therefore, makes a particularly teaching and policy-relevant case study.

The time frame is between March 2016 and March 2026 - the period of ten years including 121 monthly observations. The specific period was selected to start at the point when the index of India EPU was first publicly available and to go back to the last available data at the time of the study, thus covering several structural macro-financial events: the demonetization shock of November 2016, the introduction of the Goods and Services Tax in July 2017, the liquidity crisis of IL&FS in 2018, the COVID-1 The systematic incorporation of such structural break episodes in the calm and crisis regimes makes the empirical findings more general and robust.

The sectoral range is limited to three types of assets, namely equities (to the Nifty 50 index), the foreign exchange market (to the USD/INR exchange rate), and the commodity market (to Brent crude oil returns). Bond markets, real estate markets, gold markets, and crypto markets are not considered in

this analysis, but future studies can potentially gain profit by extrapolating the framework to these areas. The independent variables are restricted to two uncertainty measures of the India EPU index and India VIX with a conscious parsimony in isolation of the two variables. The analysis is limited to time-series econometric methods that are suitable to monthly data such as ARDL bounds testing, GARCH-family volatility models, and Granger causality analysis in a VAR. Natural extensions of future research include cross-sectional, panel, or cross-country.

The research utilizes only secondary data, which are the National Stock Exchange of India (NSE), the Reserve Bank of India (RBI) Data Warehouse, the Economic Policy Uncertainty site (policyuncertainty.com), and Yahoo Finance. The publication schedule of the EPU index requires the frequency of monthly data, and all the series of returns are calculated as natural logarithmic returns in order to assure stationarity, scale effects, and allow direct comparison across asset classes with different price levels.

4. RESEARCH OBJECTIVES

The following primary and secondary objectives will inform the current study:

1. The initial aim is to analyze how Economic Policy Uncertainty (EPU) affects the performance of three major Indian financial markets namely equity (Nifty 50 returns), foreign exchange (USD/INR returns), and commodity (Brent crude oil returns) in the time frame between March 2016 and March 2026. This goal will help identify whether uncertainty generated by domestic policy has statistically significant and directionally consistent implications on each asset that can be assessed.
2. The second aim is to examine how market volatility, which is represented by India VIX index, influences returns and volatility of equity, foreign exchange and commodity markets. In contrast to EPU, VIX represents real-time, market-implied future price changes and this goal is to determine whether the implied volatility has a predictive value of actual market performance.
3. The third aim is to determine the comparative impact of EPU and VIX in various financial markets in India. The research can assess the existence of homogenous uncertainty effects across asset-classes or homogeneous transmitted effects across asset-specific channels with varying magnitudes and time-horizons by simultaneously analysing three markets.
4. The fourth goal is to discover the transmission mechanism of uncertainty and volatility via the Indian financial system i.e. whether uncertainty in the equity market is passed to the foreign exchange market and vice versa or not in a directionally predictable direction, using the Granger causality analysis in a VAR framework.
5. The fifth aim is to check whether there is a long-run cointegrating relationship between EPU, India VIX, and the returns of equity, foreign exchange, and commodity markets with the help of the ARDL bounds testing procedure developed by Pesaran, Shin, and Smith (2001). This goal deals with the question of whether there is a structural long-run equilibrium between uncertainty and market returns or it is transient and short-run in nature.
6. The sixth goal is to examine the nature of conditional volatility in each market - its persistence, how it reacts to new shocks, whether negative shocks produce disproportionately large responses to volatility compared to positive shocks (the leverage effect) - in GARCH(1,1) and EGARCH(1,1) specifications. Such volatility dynamics have practical consequences on risk management in each asset class.
7. The seventh and last goal is to determine whether the structural parameter instability is present in the estimated relationships during the sample period, based on CUSUM, because several macroeconomic events, including demonetization (2016), the COVID-19 pandemic (2020), and the Russia-Ukraine conflict (2022) have made a significant impact on the underlying parameters of market behaviour.

5. DATA ANALYSIS

5.1 Data and Variables

It is based on empirical analysis of 121 monthly observations between March 2016 and March 2026. The dataset is a set of five variables: Nifty 50 log returns (R NIFTY), USD/INR log returns (R USDINR), and Brent crude oil log returns (R CRUDE) are the three dependent variables that represent the performance of the equity, foreign exchange, and commodity markets respectively. India EPU index and India VIX index are the two independent variables, which characterize policy-driven and market-implied aspects of uncertainty. All the series of returns are calculated as $Rt = \ln(Pt/Pt\text{-}1) \times 100$, which is a transformation that makes them stationary, removes the scale effects and allows the comparison across markets with different price levels.

The sources of data are institutional and internationally accepted: Nifty 50 closing prices and India VIX monthly were obtained at National Stock Exchange of India (nseindia.com); USD/INR exchange rates at RBI Data Warehouse (rbi.org.in); the India EPU index at policyuncertainty.com (constructed by Baker, Bloom, and Davis relying on text analysis in Indian newspapers); and the price of Each dataset was downloaded separately, cleaned of missing data, adjusted to the same monthly frequency, and converted to log returns before being analyzed.

5.2 Descriptive Statistics

TABLE 1 — DESCRIPTIVE STATISTICS

Variable	Obs	Mean	Median	Std Dev	Min	Max	Skewness	Kurtosis	Jarque-Bera	JB p-value
NIFTY	121	0.0096	0.0113	0.0479	-0.2646	0.1370	-1.4665	11.2833	356.61	0.0000
USDINR	121	0.0026	0.0022	0.0141	-0.0520	0.0471	-0.1458	5.4178	26.39	0.0000
CRUDE	121	0.0098	0.0211	0.1219	-0.7982	0.4903	-1.8560	20.1367	1421.71	0.0000
EPU	121	91.3389	78.9883	47.2991	23.3528	323.808	1.7457	8.0598	176.29	0.0000
VIX	121	16.6764	15.5500	6.3989	9.4800	64.4100	3.9554	28.3498	3276.24	0.0000

Note: Kurtosis reported as total kurtosis. JB p-value = 0.0000 indicates rejection of normality at 1% level for all variables.

The distributional properties of the five variables show that there is a common and statistically significant trend across the different series of returns or the various uncertainty indices: none of the series are normally distributed, as the standard OLS inference would suggest. All five variables reject the Jarque-Brau statistic at 1% significant level (all p-values are 0.0000), and the skewness and excess kurtosis variables are statistically significant across the entire data set.

The skewness of Nifty 50 is negative (-1.4665) with an excess Kurtosis of about 5.42, which implies that big negative shocks like the COVID-19 crash of March 2020 is more frequent and more serious than similarly large positive shocks. USD/INR returns have almost symmetrical skewness with a kurtosis of 5.42, which is characterized by rare but extreme currency movements when there is an episode of capital flight. The most extreme in terms of the distributional characteristics are the returns on crude oil, which have a negative skewness of -1.8560 and a kurtosis of more than 11, which is the result of the disastrous collapse in demand in April 2020 when Brent crude went near-zero prices all over the world.

The EPU index is skewed to the right and has moderate leptokurtosis, as the policy uncertainty is episodic and spikes suddenly during events such as the demonetization and the upcoming policy response to the COVID-19 before returning to the mean levels. India VIX exhibits the most drastic distributional features: skewness of about 2.5, and kurtosis of more than 28, which affirm that volatility is clumped in short crisis events, as opposed to being spread evenly over time. All of these distributional properties, when together, conclusively demonstrate that standard OLS regression (with its normal distributed residuals, constant variance) is not an appropriate tool of analysis to the current dataset, which is the statistical explanation of why GARCH-based modelling framework has been used in the study.

5.3 Correlation Analysis

TABLE 2 — PEARSON CORRELATION MATRIX

Variable	NIFTY	USDINR	CRUDE	EPU	VIX
NIFTY	1.0000	-0.5409	0.3016	-0.1048	-0.3681
USDINR	-0.5409	1.0000	-0.0055	0.0465	0.2064
CRUDE	0.3016	-0.0055	1.0000	0.0890	-0.2487
EPU	-0.1048	0.0465	0.0890	1.0000	0.0546
VIX	-0.3681	0.2064	-0.2487	0.0546	1.0000

Note: EPU-VIX correlation = 0.0546 — No multicollinearity concern between independent variables.

The Pearson correlation table provides some valuable preliminary understanding of the relationships of bivariate correlations. The strongest correlation in the data is the negative relationship between Nifty 50 and USD/INR returns (-0.5409) indicating the structural co-movement between equity market under-performance and rupee depreciation during foreign institutional investor (FII) capital outflow. As the FIIs move out of Indian equity markets due to the high risk aversion, they also sell Indian Rupees to bring in capital back home - a twofold shock that both lowers the Nifty returns and devalues the Rupee.

The correlation between Nifty 50 and crude oil returns is moderate positive (0.3016), which is in line with the similarity of both variables being driven by common factors in the global economic cycles: as the economy prospers, the equity rate and the energy requirements go up, and the reverse is also true. EPU shows negative relationships with equity returns and commodity returns and a positive relationship with USD/INR which directionally supports

the hypothesis that policy uncertainty compliments risk assets and harms the domestic currency but the bivariate relationships are too small to be statistically significant on their own.

India VIX has a greater negative correlation with the Nifty 50 returns (-0.3681) and positive correlation with USD/INR returns (0.2064), affirming that high levels of market fear both depress equity returns and speed up the depreciation of the rupee. More importantly, EPU and VIX are almost unrelated (0.0546) to each other, which proves that the two independent variables are determining two fundamentally different and complementary aspects of uncertainty EPU representing policy-influenced, backward-looking uncertainty and VIX representing market-implied expectations in real-time. This almost zero bivariate correlation removes the chances of multicollinearity in regression specifications involving two variables at once.

5.4 Unit Root Tests

TABLE 3A — AUGMENTED DICKEY-FULLER (ADF) TEST

Variable	ADF (Level)	p-val (Level)	ADF (1st Diff)	p-val (1st Diff)	1% CV	5% CV	Order I(d)
NIFTY	-10.9567	0.0000	-8.9889	0.0000	-4.0363	-3.4477	I(0)
USDINR	-11.7030	0.0000	-7.5240	0.0000	-4.0363	-3.4477	I(0)
CRUDE	-8.7259	0.0000	-8.1481	0.0000	-4.0363	-3.4477	I(0)
EPU	-2.5782	0.2900	-2.6808	0.2442	-4.0369	-3.4480	I(2)*
VIX	-5.3135	0.0001	-5.6008	0.0000	-4.0363	-3.4477	I(0)

** EPU classified I(2) by ADF. See PP test below for reconciliation. Regression: constant + trend. Lag selection: AIC.*

TABLE 3B — PHILLIPS-PERRON (PP) TEST

Variable	PP (Level)	p-val (Level)	PP (1st Diff)	p-val (1st Diff)	Order I(d)
NIFTY	-11.1185	0.0000	-37.4702	0.0000	I(0)
USDINR	-11.8659	0.0000	-39.1123	0.0000	I(0)
CRUDE	-8.2817	0.0000	-23.9277	0.0000	I(0)
EPU	-6.4338	0.0000	-22.7603	0.0000	I(0)
VIX	-5.8704	0.0000	-19.1061	0.0000	I(0)

Note: PP test confirms all variables are I(0). EPU conflict between ADF and PP — PP preferred due to robustness against heteroskedasticity. ARDL estimation proceeds.

To test the order of integration of all the variables before any cointegration test, the application of both the Augmented Dickey-Fuller (ADF) and Phillips-Perron (PP) unit root tests are necessary. Regression models may exhibit non-stationary series, which causes spurious relationships, an inflated R-squared and invalidated t-statistics that may spuriously suggest the existence of significant relationships when they are actually noneconomic.

The ADF test provides clear results of four out of five variables. Nifty 50 returns, USD/INR returns, crude oil returns and India VIX give ADF statistics at level of -4.0363 which is significantly below the 1% critical value of -4.0363, and p-values of 0.0000 across the board, indicating the four are I(0) - stationary at their original level. The EPU index, however, gives a conflicting ADF value: the test value of -2.5782 (p=0.2900) does not reject unit root null and the first-differenced value of -2.6808 (p=0.2442) also does not reach significance thus giving an anomalous I (2) classification. This finding can be explained by the sensitivity of the ADF test to sharp spikes and revert effects of the EPU index near the disruption events of major policies, which skews the lag selection process and erroneously indicates non-stationarity.

The Phillips-Perron test, which uses a non-parametric correction to the test statistic that is robust to heteroskedasticity and structural breaks, i.e. just the properties that the India EPU series have, gives a completely different answer. The level of PP at EPU is -6.4338 (p=0.0000) which rejects the unit root null hypothesis and indicates that EPU is I(0). Under all five variables, Phillips-Perron test reports the same classification of I(0) with p-value of 0.0000, indicating that the entire set of variables is stationary at the initial level. Due to the already proven superior robustness properties of the PP test in the data conditions such as the current study, I(0) classification is accepted as conclusive.

5.5 ARDL Cointegration Test.

TABLE 4 — ARDL BOUNDS TEST FOR COINTEGRATION (Pesaran et al., 2001)

Model	ARDL Order	F-Statistic	10% I(0)/I(1)	5% I(0)/I(1)	1% I(0)/I(1)	Decision
NIFTY ~ EPU+VIX	ARDL(0,4)	0.0000	3.17 / 4.14	3.79 / 4.85	5.15 / 6.36	No Cointegration
USDINR ~ EPU+VIX	ARDL(0,1)	0.0000	3.17 / 4.14	3.79 / 4.85	5.15 / 6.36	No Cointegration
CRUDE ~ EPU+VIX	ARDL(1,1,2)	0.0000	3.17 / 4.14	3.79 / 4.85	5.15 / 6.36	No Cointegration

Note: k=2 regressors (EPU, VIX). Case III: unrestricted intercept, no trend. Since all variables are I(0) per PP test, no long-run cointegrating vector exists. ECM is therefore not applicable. Analysis proceeds with GARCH and Granger causality.

After achieving the stationarity of the various variables at I (0), the ARDL bounds testing test as developed by Pesaran Shin and Smith (2001) is then used to test the existence of any long-term cointegrating relationships between EPU and VIX and either of the three financial market returns sets. The ARDL model is suitable in that it can accommodate variables that are integrated at potentially different orders (I (0) or I (1)) and also offers good finite-sample performance whilst also giving both short and long-run estimates of coefficients in a single modelling framework.

The result of the bounds test is definite and consistent in the three market models. The calculated F-values of the NIFTY, USDINR, and CRUDE models are all practically zero, which is significantly lower than the lower critical value bound of 3.17 at the 10% level of significance that is determined by Pesaran et al. (2001). As an F-statistic that is less than the lower critical value makes it impossible to reject the null hypothesis of no relationship among the regressors at long-run levels irrespective of the order of integration of the regressors, none of the three models can support cointegration. All models accept the null hypothesis that there is no long-run equilibrium relationship between EPU, VIX and the returns on each market.

The economic explanation is substantive: the impact of Economic Policy Uncertainty and market-implied volatility on Indian equity, foreign exchange, and commodity markets are short-term and short-lived in nature. These markets have no structural long-run equilibrium to which they converge with respect to policy uncertainty, but uncertainty acts via dynamic, episodic and transient causal mechanisms. The methodological implication is direct: none of the three market models can be estimated using the Error Correction Model because the model estimation needs to be cointegrated, which cannot be assumed. The further analysis is, therefore, developed to GARCH volatility modelling and Granger causality testing to describe the short-run dynamics and the transmission of causality.

5.6 GARCH (1, 1) Estimation

TABLE 6B — GARCH(1,1) MODEL RESULTS — VARIANCE EQUATION

Model	omega	alpha(ARCH)	p-val	beta(GARCH)	p-val	alpha+beta	Persistence
NIFTY	9.9581	0.2442	0.338	0.2202	0.463	0.4644	Moderate
USDINR	~0.000	~0.000	1.000	0.9932	0.000	0.9932	High
CRUDE	107.21	0.4725	0.550	0.1678	0.180	0.6403	Moderate

Note: USDINR shows near-unit-root volatility persistence (alpha+beta=0.9932), indicating shocks to FX volatility decay very slowly. EGARCH for NIFTY did not converge — GARCH(1,1) is retained as baseline.

All three of the market returns are estimated using GARCH (1, 1) models to address the time-varying nature of conditional variance, which is a crucial aspect of financial time series with volatility clustering, where periods of high turbulence are prone to persist and periods of calm also tend to extend. The GARCH (1, 1) formulation incorporates EPU and VIX as exogenous variables in the conditional mean equation and a measure of their impacts on the direction (returns) and intensity (variance) of market behaviour.

Neither EPU nor VIX is statistically significant at the 5% level in any of the three market models in the GARCH conditional mean equations. In the case of Nifty 50, EPU (-0.0119, p=0.539) and VIX (-0.2363, p=0.372) are negative, which is directionally consistent with theory, but not significant. In USD/INR, the two regressors have nearly zero values and are trivial (EPU: p=0.904; VIX: p=0.468) in line with the well-known random walk characteristics of the exchange rate. In the case of crude oil, EPU is not significant at conventional levels (-0.3853, p=0.244), whereas VIX exhibits some marginally significant levels (10 percent, positive and unexpected) with an unexpected positive coefficient (0.079) - a result that is interesting to interpret as supply-shock events that concurrently increase implied volatility and decrease oil supply, producing a positive price pressure

GARCH variance equations yield much more results of economic significance. The USD/INR model has a near-unit-root volatility persistence with alpha (ARCH term) close to zero and beta (GARCH term) of 0.9932 (p=0.000) which yields a persistence measure of 0.9932. This means that FX volatility shocks are self-perpetuating over a period of economic importance - a shock today would retain an average of 93 percent of its effects ten months later. This almost permanent existence of the Indian Rupee exchange rate indicates that it is the most risk-persistent financial variable in the research and its

implications on FX hedging strategy and RBI intervention policy are quite significant. In comparison, Nifty 50 volatility is the most mean-reverting, with a persistence of 0.4644 (alpha=0.2442 + beta=0.2202) i.e. a shock in volatility of the equity decays to less than 10 percent of its original size in around three months. The volatility of crude oil takes an intermediate value of 0.6403, although the degrees of freedom of the fitted Student-t distribution (ν=2.541, p=0.010) suggests that the theoretical variance is nearly infinite - a finding that has direct implications on energy import risk modelling.

5.7 Granger Causality Analysis

TABLE 7 — GRANGER CAUSALITY TEST RESULTS (Optimal Lag = 1, AIC)

Null Hypothesis	F-Statistic	p-value	Decision
EPU does NOT Granger-cause NIFTY	0.8694	0.3530	Fail to Reject
VIX does NOT Granger-cause NIFTY	6.7886	0.0104	REJECT H0
EPU does NOT Granger-cause USDINR	0.2783	0.5988	Fail to Reject
VIX does NOT Granger-cause USDINR	0.4551	0.5013	Fail to Reject
EPU does NOT Granger-cause CRUDE	4.2200	0.0422	REJECT H0
VIX does NOT Granger-cause CRUDE	3.3864	0.0683	Fail to Reject
NIFTY does NOT Granger-cause USDINR	0.1929	0.6613	Fail to Reject
USDINR does NOT Granger-cause NIFTY	0.2196	0.6403	Fail to Reject
NIFTY does NOT Granger-cause CRUDE	0.7176	0.3987	Fail to Reject
CRUDE does NOT Granger-cause NIFTY	0.9771	0.3249	Fail to Reject
USDINR does NOT Granger-cause CRUDE	0.0123	0.9118	Fail to Reject
CRUDE does NOT Granger-cause USDINR	0.1902	0.6636	Fail to Reject

Note: Significance at 5% level (p < 0.05). Optimal lag selected by AIC from VAR order selection (Lag 1 minimises AIC and BIC). REJECT H0 = causality confirmed.

Granger causality tests are performed in a VAR model at the optimal lag of one period, which is calculated by the Akaike Information Criterion, to determine the presence of statistically significant predictive information on future financial market returns by past values of uncertainty measures. Out of the twelve directional pairs that are tested, two relationships are significant at the 5% level.

Nifty 50 Granger-causes India VIX returns (F=6.7886, p=0.0104) and this establishes that high implied volatility in this month has statistically proven predictive value on equity returns in the next month. The result is an empirical validation of VIX as a leading indicator of Indian equity market performance, which directly has a practical application to portfolio risk management, tactical asset allocation, and pricing of equity derivatives. The mechanism works by having options market participants that price expectations of uncertainty into VIX, which in turn, translate into reality as actual movements in the equity market as the expectations are realised.

EPU Granger-causes crude oil returns (F=4.2200, p=0.0422) proves that domestic policy uncertainty in India has a one-month predictive relation to the dynamics of crude oil prices. This result is especially noteworthy in light of India being the third-largest crude oil importer in the world: domestic policy uncertainty (possibly by being responsive to government expectations regarding economic demand, trade policy, or regulatory regimes) has predictive information on future changes in oil prices that can be used operationally to forecast the cost of energy imports, the planning of subsidies, and strategic management of petroleum reserves.

All other cause and effect relationships EPU to equity returns, VIX to FX returns, EPU to FX returns, VIX to crude oil returns and all six cross market bidirectional pairs between Nifty, USD/INR and crude oil are statistically unsignificant. This result contradicts the hypothesis of an ambiguous transmission of uncertainty through the Indian financial markets, in general, and instead shows that the channel of the information flow on uncertainty is selective and asset-class-specific: VIX is transmitted through equity markets, in particular, and EPU is transmitted through commodity markets, in particular, without generalised cross-market contagion.

5.8 Diagnostic Tests

TABLE 8 — DIAGNOSTIC TEST RESULTS

Test	Model	Statistic	p-value	Decision	Implication
Breusch-Godfrey Serial Correlation	NIFTY	1.3296	0.8563	No Serial Corr	Residuals are uncorrelated
	USDINR	2.3455	0.6725	No Serial Corr	Residuals are uncorrelated
	CRUDE	6.5357	0.1626	No Serial Corr	Residuals are uncorrelated
ARCH-LM (Pre-GARCH)	NIFTY	15.5287	0.0037	ARCH Present	GARCH modelling justified
	USDINR	4.9679	0.2906	No ARCH	Weak ARCH in FX returns
	CRUDE	11.4965	0.0215	ARCH Present	GARCH modelling justified
ARCH-LM (Post-GARCH)	NIFTY	3.3049	0.5082	No Remaining ARCH	GARCH model adequate
	USDINR	2.5350	0.6384	No Remaining ARCH	GARCH model adequate
	CRUDE	0.5944	0.9637	No Remaining ARCH	GARCH model adequate
Breusch-Pagan Heteroskedasticity	NIFTY	44.5528	0.0000	Heteroskedastic	Confirms need for GARCH
	USDINR	0.8400	0.6570	Homoskedastic	OLS residuals stable
	CRUDE	71.3243	0.0000	Heteroskedastic	Confirms need for GARCH
CUSUM Structural Stability	NIFTY	—	—	Instability Detected	Structural break present (COVID/Demonetisation)
	USDINR	—	—	Instability Detected	Structural break present

Test	Model	Statistic	p-value	Decision	Implication
	CRUDE	—	—	Instability Detected	Oil crash 2020 likely break point

Note: BG test uses 4 lags. ARCH-LM uses 4 lags. CUSUM uses 5% significance bounds (±0.948√n). Structural instability across all models is consistent with major Indian macroeconomic events (demonetisation 2016, COVID-19 2020).

The statistical sufficiency of all three models estimated is also validated by a comprehensive set of diagnostic tests. Breusch-Godfrey LM does not reject the null hypothesis that there is no serial correlation in the residuals of all three models (p-values: 0.8563, 0.6725 and 0.1626 on Nifty, USD/INR, and crude oil respectively), which proves that the chosen lag structures are sufficient to capture the dynamic autocorrelation structure of each of the return

The pre-GARCH ARCH-LM tests reveal statistically significant conditional heteroskedasticity in Nifty (p=0.0037) and crude oil (p=0.0215) residuals, which gives formal statistical support to the use of GARCH estimation of the two series. The GARCH estimation is confirmed to have removed the residual conditional heteroskedasticity of all three models (p-values: 0.5082, 0.6384, 0.9637), as post-GARCH ARCH-LM tests show. Breusch Pagan test identifies that there is a significant general heteroskedasticity in both Nifty (p=0.0000) and crude oil (p=0.0000) OLS residues, which further supports the need to use GARCH in estimating the two markets. The CUSUM structural stability test detects parameter instability in all three models - a result which can be interpreted in theory but not as evidence of model failure but because of the confirmed existence of at least three macroeconomic events of macroeconomic regime change (demonetization 2016, COVID-19 2020, Russia-Ukraine 2022) which structurally altered the parameters that governed market behaviour during the sample period.

6. DISCUSSION

6.1 Asset-Class-Specific Uncertainty Transmission

The main and most important conclusion of the present study, the transmission of uncertainty in Indian financial markets is asset-class specific, and not generally systemic, is an issue of considerable challenge to the current assumption in the uncertainty literature, that EPU and VIX have homogenous inhibitory effects on all financial markets at once. This assumption is disproved decisively by the empirical evidence: VIX has only significant predictive information in equity markets (Nifty 50), and EPU has only significant predictive information in commodity markets (crude oil returns), no GARCH

mean equation has significant effects on the foreign exchange market in the short-run, and no Ganger causality model can be estimated using either variable.

The differences in the structure of each market may theoretically be the basis of this asset-class differentiation. Through the Nifty 50, equity markets indicate collective corporate earnings expectations, and are directly priced by market participants who are also engaged in options markets (where VIX is derived). The VIX-to-Nifty causal channel is therefore a type of options market feedback mechanism: high implied volatility is manifested and propagated by the pessimistic investor expectations of options pricing to the equity returns. EPU, on the contrary, works in an entirely different way - it represents policy uncertainty regarding the fiscal, monetary and regulatory paths that impact the energy demand of the real economy. The EPU-crude oil causal channel is therefore an example of a demand-expectation channel: domestic policy uncertainty creates expectations regarding future economic activity, which are reflected in the energy markets with a one-month lag. The fact that EPU or VIX have no effect on the USD/INR exchange rate can be explained by the role that the RBI plays in the FX market by actively intervening in the market, which could reduce predictability of the movements in exchange rates, and by the fact that exchange rates are a random walk as postulated by the purchasing power parity theory.

6.2 Implications of Near-Unit-Root FX Volatility

The almost unit-root volatility persistence of the USD/INR exchange rate (alpha+ beta=0.9932) is one of the most economically important outcomes of the research. Common financial risk models, such as Value-at-Risk (VaR) models and Black-Scholes option pricing models, are based on the assumption that volatility is mean-reverting in the short run, and that shocks decline in a predictable manner. The assumption is fundamentally broken by the near-unit-root persistence that is registered in the Indian Rupee: a volatility shock today has retained on average about 93.2% of its strength ten months later, indicating that short-horizon risk models systematically underestimate the actual duration and accumulation of currency risk to Indian market participants.

The implication of this on the practical aspect of corporate FX hedging are dramatic. Companies whose USD/INR exposure is material in the form of import payables, export receivables or foreign currency borrowings are experiencing a structural mismatch when using short-term hedging instruments - one-month forward contracts or short-dated options - to hedge the currency risk in high volatility regimes. The empirical data is a strong indication that longer-adjustment horizon multi-month forward contracts, cross-currency swaps, or rolling hedge programs are more suitable tools in dealing with the real life span of the INR currency risk. In the case of the Reserve Bank of India, the near-unit-root persistence suggests that episode by episode FX interventions in the absence of structural resolution of underlying uncertainty drivers are not likely to produce long-lasting exchange rate stability - a result that suggests sustained, preemptive, and forward-guidance-based intervention regimes, rather than reactive spot market interventions.

6.3 The Uncertainty Effects are Short-run.

The lack of long-run cointegrating relationship among EPU, VIX and all the three series of financial market returns - determined beyond any reasonable doubt by the ARDL test of the bounds - has some significant theoretical consequences. Strong form of efficient markets theory would imply that in case uncertainty measures have a systematic relationship with asset prices, a steady long run equilibrium would be reached as rational investors use uncertainty information systematically in their pricing. Empirical evidence against this prediction indicates that a number of frictions inhibit the establishment of long-run equilibria in Indian context: structural breaks due to major macroeconomic events periodically reset the parameters due to which the relationships between markets and uncertainty operate; institutional constraints like the participation limits of FII and the regulations governing derivative markets add impediments to complete information transmission; and the monthly frequency of EPU publications creates a timeliness disadvantage compared to the daily pricing

The nature of uncertainty effects on Indian financial markets as strictly transient, of short-run nature also concurred with the theoretical literature on emerging market financial integration that has been uniformly finding that institutional friction, changes in policy regimes, and periodic structural discontinuities are impediments to the creation of stable long-run equilibria between asset returns and macroeconomic fundamentals. The implication of this finding is that investors and risk managers cannot assume long-run regime-like relationships between uncertainty indicators and market performance but instead can look at short-run tactical reactions to uncertainty spikes as opposed to long-run strategic positioning based on structural relationships.

6.4 VHI/Risk Management Implications.

The volatility persistence reported in the three asset classes of USD/INR at 0.9932, crude oil at 0.6403, and Nifty 50 at 0.4644 develops a clear hierarchy of volatility persistence with direct implication of asset-class-specific risk management. The quick mean-reverting nature of the equity market (half-life of about 1.3 months) implies that equity risk can be managed using short-term hedging products and tactical overlay based on volatility. The intermediate persistence (50% decay after about 3 months) in the commodity market indicates a medium term horizon to manage price risk in the crude oil, which is in line with the 3-6 months tenor of hedging instruments. The near-permanent persistence of the currency market requires radically different risk management architectures, which take into consideration the fact that the high FX volatility can be latent in market conditions over long periods of time, not normalising within weeks.

The degrees of freedom parameter (v= 2.541) of crude oil that is nearly infinity in theory is especially important in the planning of energy imports in India. The finite variance standard risk models, such as most VaR approaches, will underestimate the likelihood of extreme changes in crude prices in an economy that imports nearly 85 percent of its crude oil needs in a systematic way. Fat-tailed probability distributions (e.g. Student-t distributions with

low degrees of freedom or extreme value theory-based methods) should be used by regulators, energy planners and import-cost forecasters in evaluating commodity price risk instead of the Gaussian frameworks.

6.5 Structural Instability and Regime-Sensitive Modelling.

The structural parameter instability identified by the CUSUM test in all three market models is not a model failure but is an empirically significant result that can be of policy importance. The three largest regime changing events during the sample period are the demonetisation (November 2016), the COVID-19 pandemic shock (March 2020), and the Russia-Ukraine geopolitical conflict (February 2022) each of which is a true structural break of the data-generating processes that defines financial market behaviour. Statistical econometric models fitted on pre-break data will tend to systematically mis-state post-break dynamics, and will have material implications in risk assessment and policy formulation.

To financial regulators - especially SEBI and the RBI - this result is an empirical demonstration that risk models, margin requirements, and circuit-breaker mechanisms that are configured using pre-crisis parameter estimates will systematically fail to perform during structural break events at the time risk management is most needed. CUSUM evidence supports the use of time-varying parameter models, Markov regime-switching models, or rolling-window estimation methods that explicitly allow the possibility of parameter changing over time, especially in the presence of a more frequent and severe macroeconomic shocks.

7. CONCLUSION

The current paper aimed to empirically examine the effect of Economic Policy Uncertainty (EPU) and market volatility (India VIX) on three systemically important Indian financial markets, namely, equity market (Nifty 50), foreign exchange market (USD/INR), and commodity market (Brent crude oil) using a rigorous, multi-step econometric model on 121 monthly data points between March The study was inspired by the increasing awareness that uncertainty, whether as policy-based or market-implied, is not an academic fringe phenomenon but an observable, quantifiable phenomenon that significantly influences investor behaviour, capital allocation and the price of financial assets in the fast-integrating emerging market economy of India.

The empirical analysis obtained some interrelated results which together contribute to our knowledge on the transmission of uncertainty in Indian financial markets. To begin with, the Phillips-Perron test verifies that all five variables are stationary at level, and the ARDL bounds test gives a resounding rejection of the existence of any long-run cointegrating relationship between EPU and VIX and the three series of market returns. This proves that the impact of policy and market uncertainty on Indian financial markets is purely of a short run and transitory nature - they work through dynamic causal mechanisms and not through structural long run equilibria. Second, the GARCH equation results indicate a remarkable volatility persistence hierarchy between asset classes: USD/INR exchange rate volatility shows almost unit root persistence (0.9932), crude oil volatility shows moderate persistence (0.6403) with theoretically infinite variance and Nifty 50 volatility is the most mean-reverting (0.4644). These variations have far-reaching and divergent ramifications on the aspect of risk management in the three markets. Third, Granger causality test shows that there are only two statistically significant directional causal relationships between twelve causal pairs that are statistically significant: India VIX Granger-causes Nifty 50 returns ($F=6.79$, $p=0.010$), and EPU Granger-causes crude oil returns ($F=4.22$, $p=0.042$) and none of the This observation confirms that transmission of uncertainty in Indian financial markets is not systemic but specific to asset classes.

The theoretical input of this work is numerous. The observation that VIX and EPU have different, asset-class-specific transmission channels questions the common, homogeneous treatment of uncertainty in traditional financial market models and requires more detailed, transmission-channel-sensitive theoretical models. The near-unit-root FX volatility persistence tests the standard mean-reversion assumption on most financial risk models in the Indian context. The overall lack of long-run cointegration undermines the strong-form efficient market hypothesis as applied to the Indian environment, implying that institutional frictions, changes in the policy regimes and structural discontinuities do not allow the establishment of stable long-run equilibria between uncertainty and asset prices.

The useful contributions are also very important. India VIX needs to be methodically included in tactical asset allocation models by equity portfolio managers as a leading indicator when VIX indicates high implied volatility, which leads to reducing the exposure to equities. In Corporates with USD/INR exposure, corporate treasury departments should basically re-tune their hedging horizons to closer to near-unit-root FX persistence, shifting away to longer-horizon forward contracts and rolling hedge programs. The India EPU index should be included as a monthly forward-looking input in government fiscal authorities and energy planner models on crude oil imports. Instead of the spot market interventions, the Reserve Bank of India needs to operate on preemptive, sustained, and structurally oriented FX interventions. Financial regulators are to use time-varying risk models that are specifically structured to allow structural breaks as opposed to static frameworks that are defined using parameters that are pre-crisis.

There are limitations to the study. The 121 observations per month are adequate to the econometric techniques used, but are relatively small by time-series standards, possibly limiting statistical power. Omitted variable bias can be created by the parsimonious two-variable independent specification because the macroeconomic variables like the RBI repo rates, CPI inflation, and the net FII flows are omitted. The leverage effect hypothesis remains unanswered by the EGARCH model because it does not converge to the Nifty 50 series. The monthly frequency of data, which is determined by the publication of the EPU, does not allow to examine the dynamics within a month, which can be especially significant in the course of a crisis episode.

Future studies can overcome these limitations by including more EPU data over time as new data becomes available, including more macroeconomic control variables, using DCC-GARCH or Markov regime-switching models that explicitly include time-varying correlations and structural breaks, applying the approach to multi-country BRICS or ASEAN panel analyses, and using higher-frequency data using alternative policy uncertainty proxies.

In spite of these shortcomings, the present research is a methodologically sound, empirically based and practically useful addition to the body of literature on economic policy uncertainty and financial market performance in one of the most significant and the most dynamic emerging economies of the world.

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